Distance and similarity

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Why is distance and similarity measurement important?

- Distance and similarity are key parts of various data mining problems where we need to compare two or more data instances. Typical examples are k-NN, kernel–based methods, or many clustering algorithms.
- Changing distance/similarity measures can lead to the change of data mining results significantly.
- Using the "*right*" similarity measure for a particular data mining problem, for a particular dataset is critical.
- There are various distance metrics and similarity measures for different types/forms of data.
- This lecture introduces basic and well–known distance and similarity measures commonly used in data mining applications.

Distance metric

Let $\mathbf{X}$ be a set or space. A distance metric on $\mathbf{X}$ is a function $d : \mathbf{X} \times \mathbf{X} \rightarrow [0, +\infty)$. For all data instances/points $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z} \in \mathbf{X}$, distance d must satisfy the following axioms:

- Identity – the distance from a point to itself is zero:

$$d(\mathbf{x}, \mathbf{x}) = 0 \tag{1}$$

- Positivity – the distance between two distinct points is always positive:

$$d(\mathbf{x}, \mathbf{y}) > 0 \tag{2}$$

- Symmetry – the distance from $\mathbf{x}$ to $\mathbf{y}$ is always the same as the distance from $\mathbf{y}$ to $\mathbf{x}$:

$$d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x}) \tag{3}$$

- Triangle inequality:

$$d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \tag{4}$$

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Minkowski distance

- Let $\mathbf{x} = (x_1, x_2 \ldots, x_m)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_m)$ be two data points in $\mathbb{R}^m$.
- The Minkowski distance between $\mathbf{x}$ and $\mathbf{y}$ is defined as follows:

$$d_r(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_r = \left(\sum_{i=1}^{m} |x_i - y_i|^r \right)^{\frac{1}{r}} \tag{5}$$

- d_r is a distance metric when $r \geq 1$. When $r < 1$, d_r violates the triangle inequality. For example, $d_r((0,0),(1,1)) > d_r((0,0),(0,1)) + d_r((0,1),(1,1))$ because $2^{\frac{1}{r}} > 1^{\frac{1}{r}} + 1^{\frac{1}{r}} = 2$ if $r < 1$.
- When $r \geq 1$, the Minkowski distance satisfies all four axioms of a distance metric.

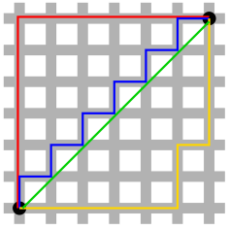
Euclidean distance

- When $r = 2$, Minkowski distance becomes Euclidean distance.

$$d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^{m} |x_i - y_i|^2} \tag{6}$$

- Euclidean distance is the length of the straight line from point $\mathbf{x}$ to point $\mathbf{y}$.
- Euclidean distance is the most well–known distance and can be used in various classification, clustering, and matching algorithms.

Manhattan distance



Euclidean and Manhattan distances [source: Wikipedia/Quora]

- When $r = 1$, Minkowski distance becomes Manhattan distance.

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sum_{i=1}^{m} |x_i - y_i| \tag{7}$$

- In the figure, the green line is Euclidean distance; the red, blue, and yellow lines are all Manhattan distance. Manhattan distance is also called *city block distance*.

Hamming distance

- Hamming distance between two numeric vectors is defined as the number of dimensions at which the their values are different.
- Hamming distance is normally used for vectors of binary or categorical values.
- Hamming distance satisfies the four axioms of a distance metric.
- For example: the hamming distance between $(1, 0, 1, 0, 1)^T$ and $(1, 1, 1, 1, 0)^T$ is 3.

Cosine distance

- Cosine of two vectors $\mathbf{x}$ and $\mathbf{y}$ is the cos of the angle θ between these vectors:

$$\cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{\sum_{i=1}^{m} x_i y_i}{\sqrt{\sum_{i=1}^{m} x_i^2}\sqrt{\sum_{i=1}^{m} y_i^2}} \tag{8}$$

- The cosine distance is defined as:

$$d(\mathbf{x},\mathbf{y}) = 1 - \cos\theta \tag{9}$$

- Cosine distance violates the identity axiom. Two **different** vectors of the same direction, e.g., $(1,1,1)$ and $(2,2,2)$, have $\cos\theta = 1$ and therefore distance $= 0$.
- Cosine distance has the value in $[0,2]$. In the positive space, i.e., $\cos\theta$ changes from 1 to 0, cosine distance has the value in $[0,1]$.
- Cosine distance/similarity is used in various data mining problems, particularly measuring the similarity between two text documents.

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Kullback–Leibler distance

- Let $P = (p_1, p_2, \ldots, p_m)$ and $Q = (q_1, q_2, \ldots, q_m)$ be two discrete probability distributions. The Kullback–Leibler distance (KL divergence) from Q to P is defined as follows:

$$D_{KL}(P\|Q) = \sum_{i=1}^{m} p_i \log \frac{p_i}{q_i} \tag{10}$$

- The above equation ignores positions where $p_i = 0$ or $q_i = 0$.
- In information theory, Kullback–Leibler is called *relative entropy*, the difference between cross–entropy $H(P, Q)$ and entropy $H(P)$.
- $D_{KL}(P\|Q) \geq 0$. $D_{KL}(P\|Q) = 0$ when the two distributions are identical. $D_{KL}(P\|Q)$ is small when the distributions have similar shapes, and large otherwise.
- Kullback–Leibler divergence satisfies the identity and positivity, but does not satisfy the symmetry axiom because $D_{KL}(P\|Q) \neq D_{KL}(Q\|P)$.

Kullback–Leibler distance (cont'd)

- In order to ensure the symmetry, we use another distribution $S = (s_1, s_2, \ldots, s_m)$ in which $s_i = \frac{p_i + q_i}{2}$. Then, the following distance is symmetric:

$$d_{KL}(P, Q) = D_{KL}(P \| S) + D_{KL}(Q \| S) \tag{11}$$

  Or using the average of both:

$$d_{KL}(P, Q) = \frac{D_{KL}(P \| Q) + D_{KL}(Q \| P)}{2} \tag{12}$$

- Kullback–Leibler distance is useful for various data mining applications. It is normally used to measure the similarity between two distributions or numeric vectors (after normalization). This distance measures the similarity at every dimension rather than only some dimensions with larger values.

Examples of Kullback–Leibler or cosine distances

- Calculate Kullback–Leibler and cosine distances between the two distributions (vectors) $P = (0.1, 0.899, 0.001)$ and $Q = (0.001, 0.95, 0.049)$:

$$
\begin{aligned}
D_{KL}(P\|Q) &= 0.1 \log\frac{0.1}{0.001} + 0.899 \log\frac{0.899}{0.95} + 0.001 \log\frac{0.001}{0.049} = 0.407 \\
D_{KL}(Q\|P) &= 0.001 \log\frac{0.001}{0.1} + 0.95 \log\frac{0.95}{0.889} + 0.049 \log\frac{0.049}{0.001} = 0.239 \\
d_{KL}(P,Q) &= \frac{D_{KL}(P\|Q) + D_{KL}(Q\|P)}{2} = \frac{0.407 + 0.239}{2} = \mathbf{0.323}
\end{aligned}
$$

- Cosine similarity:

$$
\begin{aligned}
d(P,Q) &= 1 - \cos\theta \\
&= 1 - \frac{0.1 \times 0.001 + 0.899 \times 0.95 + 0.001 \times 0.049}{\sqrt{0.1^2 + 0.899^2 + 0.001^2}\sqrt{0.001^2 + 0.95^2 + 0.049^2}} \\
&= 1 - 0.993 = \mathbf{0.007}
\end{aligned}
$$

Using Kullback–Leibler or cosine distances?

- In the previous example, the second dimension of the two vectors ($p_2 = 0.899$ and $q_2 = 0.95$) dominates the cosine similarity despite the fact that the values at the first and the second dimensions (0.1 vs. 0.001 and 0.001 vs. 0.049) are much different.
- In terms of the angle θ, the cosine similarity of 0.007 is appropriate because the θ angle between the two vectors is very small.
- However, in terms of probability distribution, these two vectors are much different. If we want to highlight this difference, KL divergence is more suitable to use.
- Therefore, KL distance is more than a distance measure, it can be seen as a constraint over every dimension when measuring the similarity.

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Similarity measure for nominal/categorical attributes

- A nominal or categorical attribute can take on two (binary) or more states. For example, gender = *male* or *female*; province = *Hanoi*, *Thai Nguyen*, *Nghe An*, etc.
- The states of a nominal attribute can be denoted by letters, symbols, or integer numbers. Notice that, integers, in this case, are used to encode states/categories. They have no meaning in ordering or comparison.
- Let $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{im})$ and $\mathbf{x}_j = (x_{j1}, x_{j2}, \ldots, x_{jm})$ be two data instances represented in m nominal attributes. The distance between the two instances is:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \frac{m - s}{m} \tag{13}$$

  where s is the number of *matches*, i.e., the number of attributes for which $\mathbf{x}_i$ and $\mathbf{x}_j$ are in the same state (same value).
- Weights for attributes can be used to highlight attributes that are more important than others.

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Similarity measure for ordinal attributes

- The values of an ordinal attribute have a meaningful order or ranking. For example, the attribute `size` has three values: *small*, *medium*, and *large*.
- Ordinal attributes may also be obtained from the discretization of numeric attributes by splitting the value range into a finite number of categories. These categories are organized into ranks.
- "*How are ordinal attributes handled*?" The treatment of ordinal attributes is quite similar to that of numeric attributes when computing distance between data instances.

Similarity measure for ordinal attributes (cont'd)

- Let an ordinal attribute X_a has K_a ordered states/values, representing the ranking $1, 2, \ldots, K_a$. Suppose a data instance $\mathbf{x}_i$ has the value x_{ia} on the attribute X_a. Replace x_{ia} by its rank, $r_{ia} \in \{1, 2, \ldots, K_a\}$.
- If we want to normalize its rank value onto [0.0, 1.0], then:

$$z_{ia} = \frac{r_{ia} - 1}{K_a - 1} \tag{14}$$

- Then, we can apply any distance metric for numeric data like Euclidean, Manhattan, cosine, etc. to the transformed ordinal attribute (using r_{ia} or its normalized value z_{ia}).

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Multivariate data revisited

- Multivariate data is a type of data which consists observations from m variables/attributes (X_1, X_2, …, X_m), i.e., considering m characteristics or attributes at the same time. This is actually a $n \times m$ matrix as follows:

$$\mathbf{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_m \\ \hline x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{pmatrix} \tag{15}$$

  In the row view, the data can be considered as a set of n data instances in the d-dimensional attribute space:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{im})^T$$

- Like univariate and bivariate data, $X_1, X_2, \ldots, X_m$ can be nominal/categorical (including binary), ordinal, or numeric.

Similarity measure for mix of nominal, ordinal, and numeric data

- Let $d(\mathbf{x}_i, \mathbf{x}_j)$ be the distance between $\mathbf{x}_i$ and $\mathbf{x}_j \in (X_1, X_2, \ldots, X_m)$. Attribute X_k $(k = 1..m)$ can be nominal, ordinal, or numeric. The distance is calculated as follows:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \frac{\sum_{k=1}^{m} \delta_{ij}^{X_k} d_{ij}^{X_k}(x_{ik}, x_{jk})}{\sum_{k=1}^{m} \delta_{ij}^{X_k}} \tag{16}$$

  where:
  - the indicator $\delta_{ij}^{X_k} = 0$ if x_{ik} or x_{jk} is missing. Otherwise, $\delta_{ij}^{X_k} = 1$
  - If X_k is numeric: $d_{ij}^{X_k}(x_{ik}, x_{jk}) = \frac{|x_{ik} - x_{jk}|}{max(X_k) - min(X_k)}$
  - If X_k is nominal or binary: $d_{ij}^{X_k}(x_{ik}, x_{jk}) = 0$ if $x_{ik} = x_{jk}$; otherwise $d_{ij}^{X_k}(x_{ik}, x_{jk}) = 1$
  - If X_k is ordinal: use the rank r_{ik} or the normalization z_{ik} and treat r_{ik} or z_{ik} as numeric.

- The distance value belongs to $[0.0, 1.0]$. We can also use $\delta_{ij}^{X_k}$ as weight of X_k that indicates how important X_k is in the distance calculation.

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Jaccard distance

- Let A and B be two sets. The Jaccard index of A and B is defined as follows:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{17}$$

- Then, the Jaccard distance between A and B is defined as:

$$d(A,B) = 1 - J(A,B) = 1 - \frac{|A \cap B|}{|A \cup B|} \tag{18}$$

- Jaccard distance satisfies all four axioms of a distance metric.
- Jaccard distance is used to measure the similarity between two sets where the order of data elements is not important.
- Different from distance on vectors, Jaccard distance can be computed between two sets of different number of elements (i.e., different in cardinality).

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Edit distance

- **Edit distance** is a well–known metric for measuring the similarity between discrete sequences like text strings, DNA sequences, etc.
- Given two sequences or strings, this distance measures the minimum number of operations required to transform one string into the other.
- This distance is widely used in text mining, natural language processing, approximate search, and bioinformatics.
- There are several variants of this distance depending on the types of operations used to transform strings. **Levenshtein distance** uses three operations: *deletion*, *insertion*, and *substitution* of a character in the string.
- Levenshtein distance is one of the most popular metric in the edit distance family and often used interchangeably with edit distance.

Levenshtein distance – algorithm

- Let $\mathbf{s}_a = a_1 a_2 \ldots a_n$ and $\mathbf{s}_b = b_1 b_2 \ldots b_m$ be two input discrete sequences or strings.
- The Levenshtein distance from $\mathbf{s}_a$ to $\mathbf{s}_b$ is the cost to transform $\mathbf{s}_a$ into $\mathbf{s}_b$ using three operations: deletion, insertion, and substitution. The distance, denoted $d_{n,m}$, is defined recursively as follows. for $i = 1..n$ and $j = 1..m$:

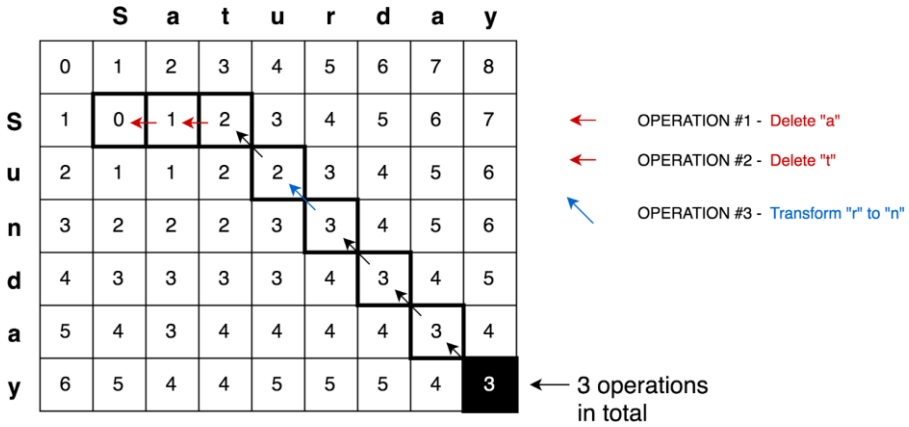
$$d_{i,0} = \sum_{k=1}^{i} w_{del}(a_k) \tag{19}$$

$$d_{0,j} = \sum_{k=1}^{j} w_{ins}(b_k) \tag{20}$$

$$d_{i,j} = \begin{cases} d_{i-1,j-1} & \text{for } a_i = b_j \\ min \begin{cases} d_{i-1,j} + w_{del}(a_i) \\ d_{i,j-1} + w_{ins}(b_j) \\ d_{i-1,j-1} + w_{sub}(a_i, b_j) \end{cases} & \text{for } a_i \neq b_j \end{cases} \tag{21}$$

In which, w_{del}, w_{ins}, and w_{sub} are the costs for deletion, insertion, and substitution.

Example of Levenshtein distance



Levenshtein distance calculation by dynamic programming [source: Internet]

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Summary

- Introduce most popular distance metrics for numeric data like Euclidean, Manhattan, cosine.
- Introduce the distance between data distributions Kullback–Leibler divergence and how this distance is different from cosine.
- Introduce the distance calculation for nominal/categorical data (including binary), ordinal data, as well as the mix of nominal, ordinal, and numeric.
- Introduce the distance measure between sets: Jaccard index and Jaccard distance.
- Introduce one of the most important similarity measure for discrete sequences or strings: edit distance and its most popular variant, Levenshtein distance.